Date change	Change log
2022-09-21	Add SHOCODE for mapping to MCID to function Generate Onepay QR code and Manual check payment status
2022-11-14	Add tag 33.03 expire on QR speciation
2022-11-14	Add Void function, add expire on Authentication request (Get JWT)
2023-10-23	Add function to generate Lao QR and Add more error message
2024-04-01	Add status on Onepay check
2026-02-05	Modify some response message and add Refund function

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1. Introduction

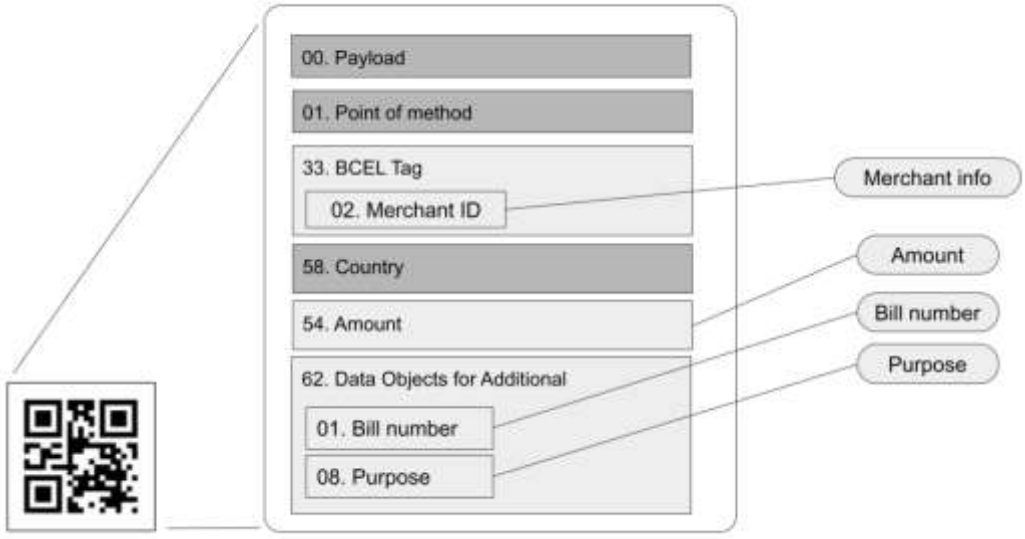
1.1. Purpose

This document outlines the API standards for third-party partners to facilitate service integration and development, covering the following key topics:

- An overview of the QR payment service workflow.
- Technical specifications for API connectivity.

1.2. QR speciation

Onepay adheres to EMVCo standards by aggregating data into structured tags to form a unified message payload. This payload is subsequently encoded into a QR code format. Each individual tag follows a specific structure comprising a Tag ID, Tag Length, and the corresponding Tag Data (Value). Per EMVCo specifications, each tag supports a maximum data capacity of 99 characters.



Onepay QR structure

tag ID	Sub tag	Field Name	Presence	format	Example
00		Payload Format Indicator	M	var up to 2	01
01		Point of Initiation Method	O	var up to 2	11
33		Allocation of Merchant Account Information	M	var up to 99	33
	00	Bank name	M	var up to 4	BCEL
	01	Service	M	var up to 8	ONEPAY
	02	Merchant ID/ Item ID	M	var up to 16	mch5949fa044ed9d
	03	Expire time	O	var up to 14	20221114083500

	05	QR Code Command	O	var up to 16	INQUIRYSTATUS
53		Transaction currency	M	var up to 3	418
54		Transaction Amount	C	number	35000
58		Country Code	M	var up to 2	LA
59		Merchant Name	M	var up to 25	Vilaivat's coffee shop
62		Data Objects for Additional	C	var up to 99	
	01	Bill Number	O	var up to 25	Aaabb112233
	05	Reference Label	O	var up to 25	
	07	Terminal Label	O	var up to 25	
	08	Purpose of Transaction	O	var up to 25	
63		Cyclic Redundancy Check		HEX	XXXX

Lao QR structure

tag ID	Sub tag	Field Name	Presence	format	Example
00		Payload Format Indicator	M	var up to 2	01
01		Point of Initiation Method	O	var up to 2	11
38		Allocation of Merchant Account Information	M	var up to 99	33
	00	Application ID	M	var up to 16	A005266284662577
	01	Bank IIN	M	var up to 8	27710418
	02	Payment type	M	var up to 3	002
	03	Merchant ID	O	var up to 16	mch5949fa044ed9d
53		Transaction currency	M	var up to 3	418
54		Transaction Amount	C	number	35000
58		Country Code	M	var up to 2	LA
59		Merchant Name	M	var up to 25	Vilaivat's coffee shop
62		Data Objects for Additional	C	var up to 99	
	01	Bill Number	O	var up to 25	Aaabb112233
	05	Reference Label	O	var up to 25	
	07	Terminal Label	O	var up to 25	
	08	Purpose of Transaction	O	var up to 25	
63		Cyclic Redundancy Check		HEX	XXXX

Example static QR code string:

00020101021133380004BCEL0106ONEPAY0216mch5949fa044ed9d52045732530341858 02LA5911 Bunny test6002VT62200716stk63719c0cafcb76304009E

QR Image:

Example dynamic QR code string:

00020101021233380004BCEL0106ONEPAY0216mch5949fa044ed9d520457325303418540520
0005802LA5910Bunny
test6003VTE625801122cb03fa4580c05122cb03fa4580c070401050814Onepay
dynamic6304EDA6

QR Image:

2. Offline QR code Generation

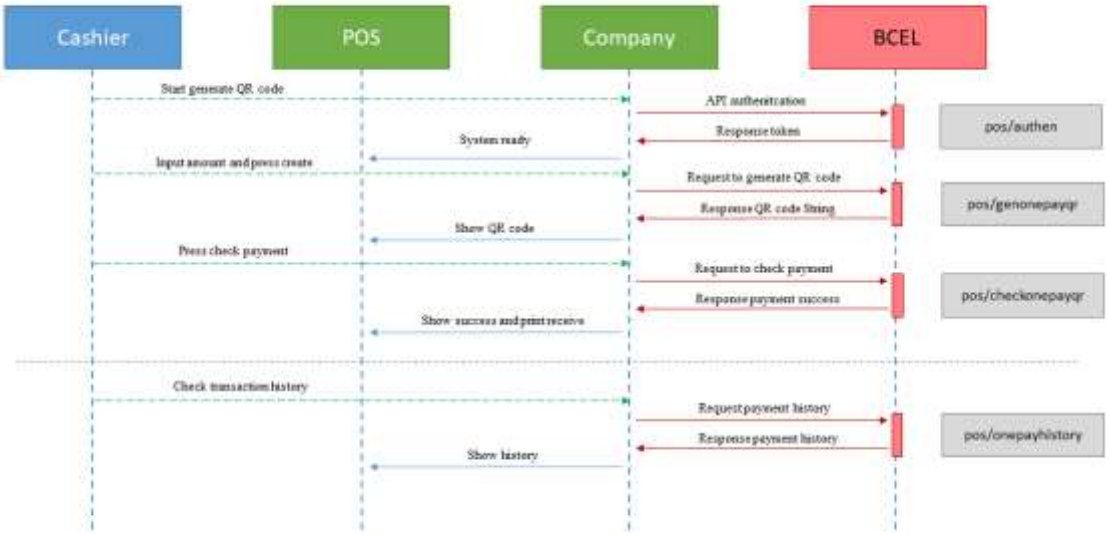
A primary advantage of this implementation is its support for offline QR generation, allowing POS terminals to generate QR codes without requiring an active internet connection. Comprehensive source code for Lao QR generation, supporting multiple programming languages, is accessible via the link provided below:

https://gitlab.com/vilaivat/bcel-lao-qr-generator

3. System Request flow

The following section outlines the sequence of request flows for the Onepay system, encompassing QR code generation, real-time payment status inquiries, and merchant transaction

history retrieval:



4. Authentication request (Get JWT)

To ensure secure access, all API requests must include a valid authentication token within the HTTP request header. This service utilizes the JSON Web Token (JWT) standard for authentication and authorization. Prior to executing any service calls, users must obtain a JWT to be included in the header of every subsequent request.

URI		**https:// bcel.la:8093/onepayservice/authen**
HTTP Method		POST
Direction	**From**	Partner
	To	BCEL

4.1. Request parameters

HTTP header

Name	Type	Manatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8

HTTP body

Name	Type	Mandatory	Example data
username	String(50)	M	Alice
password	String(100)	M	@wonderland

4.2. Sample Request and Response

Sample response success


```
{
   "user": {
      "NAME": "Vilaivat user",
      "LINKACCOUNT": "010120000042689001",
      "P_REPORT": "1",
      "P_VOID": "0",
      "P_ADMIN": "1",
      "P_SUPER": null,
      "ID": "801",
      "MCID": "mch6017b290520cf",
      "ISPROMPTPAY": "0",
      "ISUPIQR": "0",
      "ISGLN": "1",
      "UPIID": "CH6017B290520CF",
      "CHANGEPASSWORD": "0",
      "MCC": "7379",
      "MC_ENG": "Vilaivat fixcom",
      "SHOPCODE": "0cf0604175",
      "CALLBACKENABLED": "1",
      "ISECOM": "0",
      "PROVINCE_CODE": "VT",
      "CONTACT": "78383900",
      "MC_USERID": "E26CB272D5300186E043AC1003750186",
      "USERID": "801",
      "MAPPING": "0"
   },
   "expire": "2022-11-15T13:20:16Z",
   "jwt":
"eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiJ9.eyJpYXQiOjE2MTQzMjYwNDIsImp0aSI6ImI
2MWZiNTNhMzBiZDAzNWYxMTJiM2NmZjIwMDRjZjAwIiwibmJmIjoxNjE0MzI1NzQy
LCJleHAiOjE2MTQzMjkzNDIsImRhdGEiOnsiTUNJRCI6Im1jaDViNDQxNzFkN2E3MDYi
LCJJU1BST01QVFBBWSI6IjIiLCJJU1VQSVFSIjoiMSIsIlVQSUlEIjoiQ0g1QjQ0MTcxRDd
BNzA2IiwiTUNDIjoiNTk2MyIsIk1DX0VORyI6IkJ1bm55IGdhZGdldHMiLCJTSE9QQ09ER
SI6IjExMjIzMzQ0NTUiLCJQUk9WSU5DRV9DT0RFIjoiVlQiLCJDT05UQUNUIjoiNTIwM
Dg3ODUiLCJNQ19VU0VSSUQiOiI4NkM0QkIzRDM1OUE0MDVCRTA1NDAyMDgyMD
gyRDE4NCJ9fQ.kUoFwGXsJHNBdqoDxqnx8vANBSwnuYm4ZAh6vVzzWrTUta-
enGgv7Gfm5og6fl3Bh-pV4JeBMe-vt6J1RFopXFHK_-
SzXzw07vYvbaadFrur2fngbBGXcmk9JKIC4n828BzHcjgG-
UhhkRa3GKZ6umJFgN93b_6iCW6jStgIzDbW2Z-
f26xt5Ju9NFHKR5a5yXg9JvAFlzsYpVE4wTPanu_99ik3BwPnhJo1L7XPAvgtEZhtIXLz2RD
wzgVN7jEutVtS9iS_OKUs_R0IY_9FI2q0t9aqRKZ3-beZ6JG9rVpOlbrMcKhvF-
P5o6Avytq0keQ08ixraLbiwpYpOZoL7w"
}
```

Sample response fail

{
"result": 1,
"message": "invalid username or password"
}

{
"result": 1,
"message": "Merchant not found"
}

5. Onepay Service API

5.1. Generate Onepay QR code

This API is designed to facilitate the generation of Onepay QR codes for display on payment terminals or Point-of-Sale (POS) systems, enabling merchants to efficiently accept and process payments from consumers.

URI		https://bcel.la:8093/onepayservice/genonepayqr
HTTP Method		POST
Direction	**From**	Partner
	To	BCEL

5.1.1. Request parameters

HTTP header

Name	Type	Manatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8
Authorization	String	M	Bearer <jwt>

HTTP body POST value

Name	Type	Manatory	Description	Example
amount	Floats	M	Amount in LAK	10000
uuid	String	M	Running number or transaction number	6085a9f7-d076047a93
invoiceid	String(15)	M	Invoice number	b0001111245
desc	String(25)	M	description	Ice Cream match
terminalid	String(15)	O	Terminal ID	01122112442
expire	String(14)	O	QR code expire date yyyymmddhh24miss	20201031235959
shopcode	String(50)	O	Shop code to mapping Onepay's MCID	CC005

5.1.2. Sample Response Parameter

Json Response success


```
{
   "result": 0,
   "message": "Operation success",
   "version": "1.1.12",
   "data": {
      "qrc":
"00020101021238590016A00526628466257701082771041802030020316mch5949fa044ed9e
52045732530341854034995802LA5910Bunny
test6002VT62690114202503240916070514202503240916070709ONEPAYPOS0816test gen
qr code63046B90"
   }
}
```


Response fail


```
{
   "result": 1,
   "message": "Parameter Missing or invalid"
}
```

```
{
   "result": 1,
   "message": "Operation fail"
}
```

```
{
   "result": 1,
   "message": "Shopcode 11236 not found"
}
```

```
{
   "result": 1,
   "message": "Merchant not found"
}
```

```
{
   "result": 1,
   "message": "UUID is duplicate"
}
```


5.2. Generate Lao QR code

This API facilitates the generation of LaoQR codes that support both domestic interbank payments within Laos and international cross-border transactions. The service is interoperable with multiple regional networks, including Thailand, Vietnam, Cambodia, and UnionPay (China). These QR codes are optimized for display on payment terminals or Point-of-Sale (POS) systems, enabling merchants to seamlessly accept payments from both local and international consumers.

URI		https:// bcel.la:8093/onepayservice/genlaoqr
HTTP Method		POST
Direction	From	Partner
	To	BCEL

5.2.1. Request parameters

HTTP header

Name	Type	Manatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8
Authorization	String	M	Bearer <jwt>

HTTP body POST value

Name	Type	Manatory	Description	Example
amount	Floats	M	Amount in LAK	10000
uuid	String	M	Running number or transaction number	6085a9f7-d076047a93
invoiceid	String(15)	M	Invoice number	b0001111245
desc	String(25)	M	description	Ice Cream match
terminalid	String(15)	O	Terminal ID	01122112442
shopcode	String(50)	O	Shop code to mapping Onepay's MCID	CC005

5.2.2. Sample Response Parameter

Json Response success


```
{
   "result": 0,
   "message": "Operation success",
   "version": "1.1.11",
   "data": {
      "qrc":
"00020101021238590016A00526628466257701082771041802030020316mch5949fa044ed9e
52045732530341854022258O2LA6002VT6257011220231023-128051220231023-
1280705000010812test payment63045DC7"
   }
}
```


Response fail


```
{
   "result": 1,
   "message": "Parameter Missing or invalid"
}
```

```
{
   "result": 1,
   "message": "Operation fail"
}
```

```
{
```

```
    "result": 1,
    "message": "Shopcode 11236 not found"
}
```

```
{
    "result": 1,
    "message": "Merchant not found"
}
```

```
{
    "result": 1,
    "message": "UUID is duplicate"
}
```


5.3. Manual check payment status

This API is designed to facilitate the verification of payment transactions initiated by consumers. It enables merchants to perform real-time status inquiries and retrieve comprehensive transaction metadata, including payment confirmation and detailed transaction information.

URI		https://bcel.la:8093/onepayservice/checkonepayqr
HTTP Method		POST
Direction	**From**	Partner
	To	BCEL

5.3.1. Request parameters

HTTP header

Name	Type	Minatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8
Authorization	String	M	Bearer **<jwt>**

HTTP body

Name	Type	Manatory	Description	Example
uuid	string	M	Unique number from generate onepay QR step	6085a9f7-d076047a93
shopcode	String(50)	O	Shopcode to mapping Onepay's MCID	CC005

5.3.2. Sample response parameter

Response success Same bank


```
{
    "result": 0,
    "message": "Operation success",
    "version": "1.1.12",
    "mcid": "mch5949fa044ed9e",
```

```
   "SERVICE": "ONEPAY",
   "data": {
      "TXTIME": "05/02/2026 15:41:44",
      "DATETIME": "2026-02-05 15:41:44",
      "NAME": "Vilayvath Moungkhounphaisane",
      "PHONE": "96764***",
      "TICKET": "7SV4I6TPCWUJ",
      "FCCREF": "202602051508802",
      "AMOUNT": "599",
      "CCY": "LAK",
      "LAKAMOUNT": "599",
      "TID": "ONEPAYPOS",
      "IID": "20250324091604",
      "UUID": "20250324091604",
      "DESCRIPTION": "test gen qr code",
      "SERVICE": "ONEPAY",
      "FROMBANK": null
   }
}
```


Response success Cross bank


```
{
   "result": 0,
   "message": "Operation success",
   "version": "1.1.12",
   "mcid": "mch5949fa044ed9e",
   "SERVICE": "LAOQR",
   "data": {
      "TXTIME": "05/02/2026 15:47:13",
      "DATETIME": "2026-02-05 15:47:13",
      "NAME": " ",
      "PHONE": null,
      "TICKET": "ULQWWK3GU9L9EFO7SKPL",
      "FCCREF": "202602053584612",
      "AMOUNT": "499",
      "CCY": "LAK",
      "LAKAMOUNT": "499",
      "TID": "ONEPAYPOS",
      "IID": "20250324091607",
      "UUID": "20250324091607",
      "DESCRIPTION": "test gen qr code",
      "SERVICE": "LAOQR",
      "FROMBANK": "KBANK"
   }
}
```


Response fail


```
{
   "result": 1,
```

```
   "message": "Parameter Missing or Invalid",
   "version": "1.1.12"
}
```

```
{
   "result": 2,
   "message": "Payment not success or Transaction not found (Can check items 3 days in the past.)",
   "SERVICE": "UNKNOW",
   "version": "1.1.12"
}
```

```
{
   "result": 1,
   "message": "Shopcode [CC005] not found"
}
```

```
{
   "result": 2,
   "message": "Generated QR",
   "SERVICE": "ONEPAY",
   "version": "1.1.12",
   "data": {
      "STATUCODE": "0",
      "STATUS": "Generated QR"
   }
}
```

```
{
   "result": 2,
   "message": "Scanned",
   "SERVICE": "ONEPAY",
   "version": "1.1.12",
   "data": {
      "STATUCODE": "1",
      "STATUS": "Scanned"
   }
}
```


5.4. Get Onepay payment history

This API facilitates the retrieval of historical Onepay transaction data, encompassing both domestic and interbank payment records. It is designed to provide merchants with comprehensive access to their transaction logs for auditing, reporting, and reconciliation purposes.

URI		https://bcel.la:8093/onepayservice/onepayhistory
HTTP Method		POST
Direction	**From**	Partner
	To	BCEL

5.4.1. Request parameters

HTTP header

Name	Type	Manatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8

Authorization	String	M	Bearer <jwt>

HTTP body

Name	Type	Manatory	Description	Example
datefrom	string	M	yyyymmddhh24miss	20201106000000
dateto	string	m	yyyymmddhh24miss	20201106235959

5.4.2. Sample Response Parameter

Response success


```
{
   "result": 0,
   "message": "Operation success",
   "version": "1.1.8",
   "data": [
      {
         "TXTIME": "2021-02-10 16:26:17",
         "MCID": "mch5b44171d7a706",
         "MC_NAME": "Bunny gadgets",
         "LAKAMOUNT": "10000",
         "PAYMENT_USERID": "04BF6AE92BF5002CE053AC100375002C",
         "MASKACCOUNT": "01012000*****26001",
         "ACCOUNT": "010120001012026001",
         "MASKNAME": "Khong Tha*****",
         "TICKET": "G046F5N9D3Z0",
         "FCCREF": "010FTBP21041E2C7"
      }
   ]
}
```


Response on not exist


```
{
   "result": 1,
   "message": "Transaction not found on target date",
   "version": "1.1.8"
}
```

```
{
   "result": 1,
   "message": "Parameter Missing or Invalid"
}
```


5.5. Set QR Scan status

This API is specifically designed for vending machine integration to manage and update the operational status of static QR codes. It enables the system to determine whether a QR code is currently active and ready for consumer scanning or if it should be temporarily disabled based on the machine's real-time availability.

URI		https://bcel.la:8093/onepayservice/setqrstatus
HTTP Method		POST
Direction	From	Partner
	To	BCEL

5.5.1. Request parameters

HTTP header

Name	Type	Manatory	example
Accept	String	M	Application/json
Content-type	String	M	Application/json;charset=utf-8
Authorization	String	M	Bearer <jwt>

HTTP body

Name	Type	presence	Description	Example
id	string	M	Fix item number from BCEL	itm6201d81e69166
amount	string	O	Amount in LAK	100
uuid	string	O	Alphanumeric Special	6085a9f7-d076047a93
isready	number	M	1: ready 0: not ready	1

5.5.2. Sample Response Parameter

Response success


```
{
   "result": 0,
   "message": "Operation success",
   "version": "1.1.10"
}
```


Response on not exist

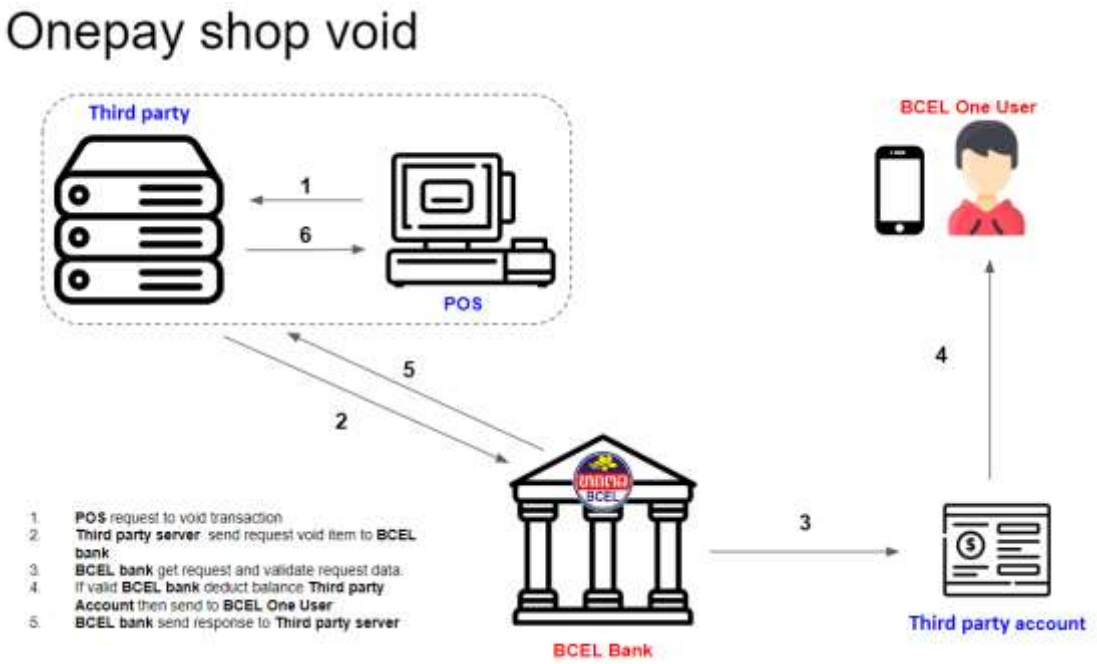

```
{
   "result": 1,
   "message": "UUID is duplicate. Please use another UUID",
   "version": "1.1.10"
}
```

```
{
   "result": 1,
   "message": "This member is not exist"
}
```

```
{
   "result": 1,
   "message": "Parameter Missing or Invalid"
}
```


6. Void function

This API provides the functionality to process refunds for consumers in cases of erroneous payments. Additionally, it supports transaction voidance, allowing merchants to cancel or nullify a transaction if the request is initiated within the same business day as the original transaction.



6.1. Generate Key exchange

To generate key third party follow below step

- Generate RSA public-private key pair, 2048 bits. E.g.:
  # openssl genrsa -des3 -out keypair.pem 2048
- Export public key in PEM format. E.g.:
  # openssl rsa -in keypair.pem -outform PEM -pubout -out public.pem
- Submit public key to BCEL.

6.2. Void request

URI		https://bcel.la:8093/onepay/void.php
HTTP Method		POST
Direction	From	Partner
	To	BCEL

6.2.1. Request parameters

HTTP body

Name	Type	Mandatory	Example data
mcid	String(20)	M	mch5b44171d7a706
uuid	String(100)	M	20220606292787
Signature *	String(100)	M	UYDck+tLLyi5D9H.....M+mT3ZIew86EhA== Signature = BASE64 (SHA256 (mcid + uuid))

6.2.2. Sample Request and Response

Sample response success


```
{
   "result": 0,
   "message": "Operation success"
}
```


Sample response fail


```
{
   "result": 41,
   "message": "Transaction not found"
}
```

```
{
   "result": 42,
   "message": "Transaction too old"
}
```

```
{
   "result": 43,
   "message": "Transaction already void"
}
```


6.3. Refund request

URI		https://bcel.la:8083/onepay/refund.php
HTTP Method		POST
Direction	From	Partner
	To	BCEL

6.3.1. Request parameters

HTTP body

Name	Type	Mandatory	Example data
mcid	String(20)	M	mch5b44171d7a706
uuid	String(100)	M	20220606292787
refundid	String	M	Refund ID generate by merchant
Signature *	String(100)	M	UYDck+tLLyi5D9H.....M+mT3ZIew86EhA== Signature = BASE64 (SHA256 (mcid + uuid))

6.3.2. Sample Request and Response

Sample response success


```
{
   "result": 0,
   "message": "Operation success"
}
```

Sample response fail

```
{
   "result": 41,
   "message": "Transaction not found"
}
```

```
{
   "result": 42,
   "message": "Transaction too old"
}
```

```
{
   "result": 43,
   "message": "Transaction already void"
}
```